

COLLECTED OBSERVATIONS:

Observations-1:

1. runtime, fuel_consumption and obddistance has similar graphs and can be combined to form a new single feature mileage which is similar to fuel_economy.
2. drivers_demand_engine_torque_percent and accelerator_pedal_pos has similar graphs and can be combined to form a new feature driver_input_demand
3. fuel_level and fl_level are same thing and one can be omitted
4. engine_torque_percent and engineload has similar graphs and can be combined to form a new feature performance_demand

Observations-2:

1. brake_switch_status is mostly released
2. clutch_switch_status is mostly pressed
3. parking_switch_status is mostly released

Observations-3:

1. The graph shows there are 2 groups of data, so we need to separate them into 2 groups and then analyse

Observations-4:

1. Now we find a relation between vehicle speed and fuel economy.
2. As we increase speed, the fuel economy also increases until speed of 70km/h.

Inference:

1. At a speed of 65-70km/h the performance of vehicle is at the highest

Observations-5:

1. fuel economy decreases as engine load increases, highest around 10-15
2. fuel economy increases as coolant increases, highest around 82-92

3. fuel economy increases as vehicle speed increases, highest around 65-70
4. fuel economy is first increasing, highest around 800, then decreasing as rpm increases
5. fuel economy decreases as engine torque percent increases, high around 5-10
6. fuel economy increases as current gear increases, high around 16-17
7. fuel rate decreases as fuel economy increases
8. fuel economy decreases as accelerator pedal position increases
9. fuel economy is highest when engine throttle valve position is near 100
10. fuel economy decreases as drivers demand for engine torque percent highest at t. 11. highest at 0
11. fuel economy has higher values at 3 modes of engine torque at 1 and 9

Observations-6:

1. adblue level is considerably at higher level when coolant is below 40
2. adblue level is considerably at higher levels when distance is below 6.3
3. adblue level is considerably at higher levels when f level is higher than 80
4. adblue level is considerably at higher levels when fuel economy is around 60-120

Observation-7:

1. adblue level is considerably at higher levels when rpm is between 50-750

CONCLUSIONS AND RECOMMENDATIONS:

Conclusion-1: Feature Engineering and Redundancy

1. Combining Similar Features:

--runtime, fuel_consumption, and obddistance have similar graphs and can be combined into a new feature mileage, representing vehicle efficiency.

--drivers_demand_engine_torque_percent and accelerator_pedal_pos can be merged into driver_input_demand to simplify the representation of driver control.

--engine_torque_percent and engine_load can be merged into performance_demand to represent overall engine workload.

2. Removing Redundant Features:

--As fuel_level and fl_level are almost identical, one of them should be omitted to avoid redundancy.

Conclusion-2: Switch Status Insights

--Given their static nature, these switch statuses might not provide significant variation for predictive models. Consider their relevance in analysis or discard if unimpactful.

Conclusion-3: Clustering for Segmentation

--The data exhibits two distinct groups, indicating potential clusters. These groups could correspond to different vehicle operating conditions or driver behaviors.

Conclusion-4: Speed and Fuel Economy Relationship

--Optimize driving strategies and algorithms to maintain speeds within the range of 65-70 km/h for enhanced fuel efficiency and performance.

Conclusion-5: Factors Influencing Fuel Economy that in turn tells about vehicle performance

--Optimize vehicle operation around the identified high fuel economy ranges and minimize conditions leading to decreased efficiency.

Conclusion-6: AdBlue Level Analysis

--Monitor and control AdBlue levels under these conditions to maintain emission standards and system efficiency.

Conclusion-7: RPM and AdBlue Levels

--Investigate why AdBlue consumption is higher at low RPM and explore strategies to optimize its use and reduce harmful emissions